

Fertility and Housing Tenure Choice

Tommaso De Santo

08/07/2026

Since May

One-market lifecycle model of housing and fertility. What changed:

- **Sequential fertility:** birth-by-birth attempts under declining fertility; children ever born and children at home tracked separately
- **Idiosyncratic income risk:** persistent after-tax earnings process, set externally
- **One housing market:** the two-location structure is set aside for the national question

Model

Environment

- Households live J four-year periods from age 18; retirement, then survival risk
- Two child states: n children ever born, m children currently at home
 - Children at home raise consumption and housing needs
 - Each child matures stochastically, leaves the household, and enters the economy as a young adult
- Idiosyncratic persistent earnings risk; wages and the interest rate are exogenous
- Households rent, or own a discrete house size on a ladder financed with a mortgage
- Government: property tax rebated lump sum; payroll tax funds a pension
- New households enter as young renters: matured children plus an outside inflow

Preferences

Flow utility over consumption and housing services:

$$u(c, \mathbf{h}; m) = \frac{1}{1 - \sigma} \left[\frac{c^\alpha (\mathbf{h} - \underline{h} m)^{1-\alpha}}{e(m)} \right]^{1-\sigma} + \psi_{\text{child}} m.$$

- **Space requirement.** Each child at home absorbs $\underline{h}$ rooms before housing yields services
- **Equivalence scale.** $e(0) = 1$ and $e' > 0$: a larger family needs more resources to reach the same utility
- **Owner premium.** $\mathbf{h} = \chi h$ for owners, with $\chi > 1$; $\mathbf{h} = h^R$ for renters

Bequests

Warm-glow utility at death over terminal wealth:

$$\mathcal{B}(\tilde{b}) = \theta_0 \frac{(\theta_1 + \tilde{b})^{1-\sigma}}{1-\sigma}, \quad \tilde{b} = b + ph.$$

- θ_0 governs the strength of the bequest motive; θ_1 makes bequests a luxury
- Estates are valued by parents but transfer to no one: no intergenerational wealth flow

Fertility and Child Maturation

- Between ages 18 and 45, households decide each period whether to try for a child: a discrete choice with taste shocks, with separate scales κ_f before and after the first child
- Trying yields a birth with the conception probability

$$\pi(\text{age}) = 1 - \omega_1 e^{\omega_2(\text{age}-18)}, \quad \pi = 0 \text{ after } 45,$$

which falls from 0.98 at 18 to 0.62 at 40: postponed births are increasingly likely never to happen

- Each child lives at home for 18 years on average, with stochastic maturation, raising the equivalence scale and the space requirement
- Matured children enter the economy as young households

Firms and Earnings

- A competitive firm produces with labor, so the wage is pinned down by technology and the labor market clears at any aggregate efficiency units L :

$$Y = A L, \quad w = A$$

- Households supply efficiency units inelastically: an age profile times a persistent idiosyncratic component

$$\log y' = \rho \log y + \varepsilon', \quad \varepsilon' \sim N(0, \sigma_\varepsilon^2)$$

- Earnings are taxed progressively, a payroll tax funds a stationary pension, and a property tax on owners is rebated lump sum
- With constant returns, a larger population leaves the wage unchanged: housing is the only fixed factor

Housing, Tenure, and Budget Constraints

Households rent a continuous quantity up to a cap, or own a discrete size on a ladder:

$$h^R \in (0, \bar{h}^R], \quad h \in \{H_1, \dots, H_K\}.$$

Tenure segmentation: $\max H_k > \bar{h}^R$, so family-sized space is reachable only through ownership.

Renters

$$c + b' + r h^R = (1 + q) b + y(j), \quad b' \geq 0.$$

Owners, with prior stock h_{-1}

$$c + b' + (\delta + \tau_H) ph + \mathbf{1}\{h \neq h_{-1}\} [ph - (1 - \psi) ph_{-1}] = (1 + q) b + y(j), \quad b' \geq -\phi ph.$$

- q : interest rate; δ : maintenance; τ_H : property tax; ψ : transaction cost; $1 - \phi$: down-payment share

Housing Supply and Market Clearing

Housing supply slopes upward in levels, and rents map into prices through user cost:

$$H^S = H_0 r^\xi, \quad r = (q + \delta + \tau_H) p.$$

- One market-clearing condition: aggregate housing services demanded equal H^S
- Supply is in levels, not per household: a permanently larger population permanently raises rents
- Stationary equilibrium: the price, the transfer, and the distribution are constant, and the government budget clears

The Household's Problem

Within the period, a household first decides whether to try for a child, then chooses tenure, housing, and saving:

$$V(b, h, y, a, n, m) = \mathbb{E}_\varepsilon \max_{f \in \{0,1\}} \left\{ W^f(b, h, y, a, n, m) + \varepsilon_f \right\},$$

$$W^f = \max_{\text{rent/own}, h', h^{R'}, b'} u(c, \mathbf{h}; m) + \beta \mathbb{E}[V(b', h', y', a + 1, n', m')]$$

subject to the renter or owner budget and the collateral constraint; a successful try ($f = 1$, probability $\pi(a)$) adds a child next period.

In old age there are no fertility decisions; survival risk and the warm glow at death close the problem.

Population Closure

Closing the Population

Every version of this model needs a rule for the size of the entering cohort. Three ways to supply one:

1. **Balanced growth path.** Population and all quantities grow at a common rate; the model is read per capita
 - Would need to handle two technical issues here: the committed housing requirement makes preferences non-homothetic, and the housing stock is fixed in levels
 - Counterfactual anyway: the economy is not observed growing along one
2. **Dynamic equilibrium path.** The most general option: no stationarity, solve for the whole time path of the population and prices, markets clearing every period
 - Expensive, and calibration becomes ambiguous: every observed moment must be assigned a date on the path
3. **Stationary equilibrium.** Compare stationary states across policies
 - One long-run number per policy; calibration targets a stationary cross section
 - Needs a rule that pins the level of the population

The stationary comparison is used here.

The Entry Rule

Entrants: matured children who stay, plus an inflow M from outside (immigration). M needs a rule:

$$\text{earlier: } M = M(V) \qquad \text{now: } M = \bar{M}$$

- **Earlier.** Entry rises with the value V of living in the economy. No moment disciplines that response, and it drives the results: population responses three to ten times larger
- **Now.** A fixed quota: policy moves the population only through births
- **Cost.** Migration cannot respond to a more attractive economy; the earlier numbers bound that channel

Population Balance

The economy has a mass S of households: the population level. In a stationary state entrants replace exits:

$$\underbrace{S E_0}_{\text{entrants needed}} = \underbrace{R S B_0}_{\text{matured children who stay}} + M,$$

with E_0 exits and B_0 maturing children, per household; R the share who stay.

- The native terms scale with S ; the inflow M is a level
- As S falls, M covers a growing share of entrants and the decline stops:
$$S = M / (E_0 - R B_0)$$
- The outside share of entrants is measured across metro areas in the ACS: 16.9% of young entrant households arrive from outside, in line with national net migration relative to cohort size. Since $s^{out} = 1 - R B_0 / E_0$, natives replace 83% of each cohort, and M sets the level

Population in Counterfactuals

R and M stay fixed, so policy moves the population through births:

$$\% \Delta S = \frac{1 - s^{out}}{s^{out}} \% \Delta B_0 = \frac{0.831}{0.169} \% \Delta B_0 \approx 5 \% \Delta B_0.$$

- Extra births become native entrants, who have children in turn; the multiplier sums the generations
- Rents push back: more households raise rents, higher rents discourage births. The offset is small

Calibration

Externally Calibrated Parameters

	Object	Value	Source
<i>Pref.</i>	σ risk aversion	2.0	standard
<i>Fin.</i>	q interest (annual)	0.04	standard
	δ depreciation (annual)	0.02	Greaney et al.
	τ_H property tax (annual)	0.01	US average
	ϕ financed share	0.80	20% down payment
	ψ transaction cost	0.06	Greaney et al.
<i>Life</i>	17 four-year periods	ages 18–85	retirement 65
<i>Income</i>	$\rho, \sigma_\varepsilon, \text{HSV } \tau$	0.914, 0.169, 0.181	FL $\times$ HSV
	τ_{pay} payroll	0.179	Greaney et al.
<i>Children</i>	maturation hazard	2/9 per period	18 expected years
	fecundity ω_1, ω_2	0.02, 0.134	conception schedule
<i>Housing</i>	ladder, rental cap	$\{2, \dots, 11\}$, 6 rooms	room units

Internally Calibrated Parameters

Param	Description	Estimate
β (annual)	Discount factor	0.9946
ψ_{child}	Child flow utility	0.000
κ_f	Attempt taste scale, first birth	2.987
κ_f^{cont}	Attempt taste scale, later births	1.713
χ	Owner service premium	1.037
H_0	Housing supply intercept	61.706
θ_0	Bequest strength	0.159
θ_1	Bequest shifter	0.094
$\underline{h}$	Child room floor	0.614

Identification

The nine parameters are estimated jointly; four economic blocks, each with the moments most informative about it:

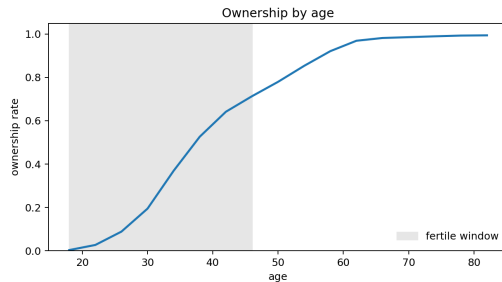
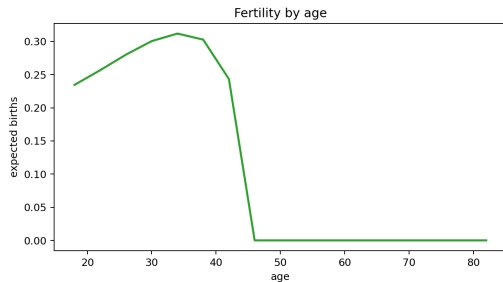
- **Fertility.** The child utility ψ_{child} raises the return to every birth: completed fertility. The first-birth scale κ_f spreads when women start: mean age at first birth and first births after 30. The continuation scale κ_f^{cont} sets how much of a commitment starting is: childlessness
- **Family housing.** The child room floor $\underline{h}$ sets the space children require: rooms added at the first birth, and the rooms gap between large and small families. The family ownership gap has no parameter of its own and overidentifies the block
- **Tenure and supply.** The owner premium χ : the ownership rate. The supply intercept H_0 : aggregate rooms
- **Saving and bequests.** The discount factor β : wealth over earnings. Bequest strength θ_0 : the bequest flow. Bequest curvature θ_1 : old-age wealth dispersion

Model Fit

	Moment	Target	Model
<i>Fertility</i>	Completed fertility	1.918	1.967
	Childlessness	0.188	0.104
	Mean age at first birth	25.31	25.57
	First births at 30+	0.270	0.327
<i>Housing</i>	Rooms response to first birth	0.664	0.690
	Rooms, 3+ vs. 1-2 children	0.368	0.359
	Family ownership gap	0.168	0.162
	Ownership rate	0.575	0.582
	Aggregate mean rooms	5.780	6.478
<i>Wealth</i>	Wealth / earnings	6.873	5.794
	Bequest flow / wealth	0.0088	0.0087
	Old wealth p90/p50	3.448	2.090

The remaining misses are childlessness, the dispersion of old-age wealth, and the share of late first births; additional search does not improve them.

Lifecycle Profiles



Policy Results

Funded Property-Tax and Grant Reforms

The experiment: raise the annual property tax from 1% to 2% of house value and return the revenue to households, either as an equal lump-sum rebate or as a grant paid at the purchase of a family-sized home. Percent changes across stationary states:

Policy	TFR	Births	Households	House price	Rent
2% tax, lump-sum rebate	+0.38	+0.33	+1.70	-17.5	-6.1
2% tax, purchase grant	+0.75	+0.67	+3.46	-17.3	-5.8

- The tax lowers the house price, so the down payment falls by 17.5%; rents fall only 6%, because rent includes the tax itself
- In U.S. units, +0.33 to +0.67% of births is roughly 12 to 24 thousand additional babies per year